NONLINEAR DISCRETE CONTROL OF A SYNCHRONOUS GENERATOR

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Abstract

In this paper we consider the problem of nonlinear discrete control of a synchronous generator. A discrete model isd eveloped and used to implement recent results on the design of linearizing control.

Keywords

Discrete system, nonlinear controller, synchronous generator, discrete control, Kronecker product.

1.Introduction

The problem of control scheme implementation in continuous time via a calculator, has attracted a great attention of authors in the area of research [9,13,14]. This problem is represented by the difficulties of generating the same performances of the controller design in continuous time at all sampling instants [6,9,13,14] .For reducing these difficulties, we are interested to study discrete time models in order to characterize control laws which offer stabilizing performance of this class of system.

Our contribution concerns the class of nonlinear analytic affine in the control discrete systems. The work that we consider in this paper allows the determination of discrete time control design, which can be implemented directly for the control of the discrete model of a synchronous generator. Our first study focuses on the determination of a discrete model of the generator. Secondly, it exploits recent results concerning the nonlinear discrete control of polynomial process.

The polynomial continuous model of the synchronous generator considered in this work is presented in [1,2,3,7,8,15]. It's an improved reduced-order, three dimensional model characterized by the following state equation:

$$\dot{\xi} = \Phi(\xi) + Gu_f \qquad (1)$$

where $\xi = [\xi_1, \xi_2, \xi_3]^T = [\delta, \omega, i_f]^T$ is a state vector with

δ : the angle between the q axes and the power system voltage U_s (generator angle).

ω: the rotor speed.

i_f : field winding current.

u_f : field winding voltage.

The state equations defining this model are given by :

$$\begin{cases}\dot{\xi}_1 = \omega_b(\xi_2 - \omega_s)\\ \dot{\xi}_2 = d_2(\xi_2 - \omega_s)\cos^2(\xi_1) + p_1\xi_3\sin(\xi_1) + p_2\sin(2\xi_1) + p_3P_{mec}\\ \dot{\xi}_3 = p_4(\xi_2 - \omega_s)\sin(\xi_1) + p_5\xi_3 + p_6u_f\end{cases} \qquad (2)$$

where the parameters p_i $i = 1, \ldots, 6$;

P_{mec}, d_2, ω_b and ω_s are detailed in appendix A and C.

This differential equations system is exploited to determine a polynomial discrete time model of the generator. Thes tudy presented in this paper includes the two following sections: First; we remind after this introduction, the polynomial nonlinear model of the synchronous generator. Then, we present the discretization approach of this continuous model. In the second section, our research will be reserved to the discrete linearizing control, where the control law and the nonlinear transformation leading to the equivalent linear model of the generator are detailed.T he effectiveness of both discrete model and digital controller are tested by means of numerical simulation study which achieve this work.

2.Discrete modeling of the synchronous generator

Let's consider the continuous model described by system (2). The development of such system in a neighborhood of an operating point (ξ_e, u_{fe}) leads to the non linear polynomial model given by :

$$\dot{x} = F_1 x + F_2 x^{[2]} + F_3 x^{[3]} + Gu \quad (3)$$

where :

$$\begin{cases} x = \xi - \xi_e \\ u = u_f - u_{fe} \end{cases} ; F_1 = \begin{bmatrix} 0 & \omega & 0 \\ a_1 & a_2 & a_3 \\ 0 & b_1 & b_2 \end{bmatrix};$$

$$F_2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ a_4 & a_5 & a_6 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & b_3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} F_3(2,1) = a_7 \\ F_3(2,2) = a_8 \\ F_3(2,3) = a_9 \\ F_3(3,2) = b_4 \\ F_3(i,j) = 0 \; for \; others \; i,j \end{cases} ; G = [0 \; 0 \; b]^T$$

parameters $a_j, j = 1...9$; b_k, $k = 1...4$ are listed in appendix B

In the following section; we consider the exploitation of system (3) in order to determine a discrete model representing as well the behavior of the synchronous generator.

Considering a suitable sampling period T, the principle of this discretization approach consists in considering these following approximations in the interval $[kT, (k+1)T[$:

$$\begin{cases} \dot{x} \approx \dfrac{x_{k+1} - x_k}{T} \\ x \approx \dfrac{x_{k+1} + x_k}{2} \end{cases} \quad (4)$$

which will be exploited in equation (3). Then one can easily write the nonlinear discrete-time model given by :

$$x_{k+1} = A_1 x_k + A_2 x_k^{[2]} + A_3 x_k^{[3]} + B u_k \quad (5)$$

where :

$$A_1 = (I_n - \overline{F}_1)^{-1}(I_n + \overline{F}_1),$$

$$A_2 = T(I_n - \overline{F}_1)^{-1}.F_2,$$

$$A_3 = T(I_n - \overline{F}_1)^{-1}.F_3, B = T(I_n - \overline{F}_1)^{-1}.G$$

and $\overline{F}_1 = \dfrac{TF_1}{2}$.

Since the ultimate goal is to check the validity and the effectiveness of the proposed discrete model, a simulation study of both systems (3) and (5) is done for the open-loop system.

The numerical parameters characterizing the machine are given in appendix C and the sampling period is chosen such that $: T = \dfrac{T_0}{10}$, where T_0 ist he oscillation period of the continuous model. The considered perturbation is a sudden variation of +50%on the nominal field winding current value. Simulation results are shown in figures (1), where one can easily notice the good agreement between variables x ofe quation (3) and x_k of equation (5). Thus the obtained discrete model (5) offers a good approximation the continuous model (3).

In the other hand; we can also establish that the system dynamic response, provides very weakly damped oscillations. Exploiting the strategy of polynomial linearizing control in the next section , we will show the advantage ofs uch strategy on the global behavioro f the generator variable.

3.Discrete linearizing controller of the synchronous generator

The study that wep resent in this paragraph consists in the exploitation of the model (5) and the nonlinear approach of discrete control proposed in [11], in order to define a discrete control law which improvest he performances of the studied generator.

3.1. Control law and non linear transformation

In this paragraph we are interested by the class of system represented by state equations of the form :

$$x_{k+1} = f(x_k) + g(x_k) u_k \quad (6)$$

where the functions $f(.)$ et $g(.)$ are supposed to be analytic.

The principle of our control linearising approach consists in the determination , when it exists, ofn onlinear feedback control law for system (6) :

$$u_k = h(x_k) = \sum_{i \geq 1} \alpha_i x_k^{[i]} \quad (7)$$

such that there exists a nonlinear state transformation :

$$z_k = T(x_k) = \sum_{j \geq 1} T_j x_k^{[j]} \quad (8)$$

where :

$$z_{k+1} = M z_k \quad (9)$$

is an arbitrary linear system representing the desired performances of the controlled system (6).

By using Kronecker product device[5,12], and by replacing u_k by its expression (7), equation (6) could be easilyw ritten in the following form :

$$x_{k+1} = \sum_{i \geq 1} F_i^1 x_k^{[i]} \qquad \textbf{(10)}$$

where $F_i^1 = f_i + \sum_{j=0}^{i-1} g_j(\alpha_{i-j} \otimes I_{n^j})$.

The z_k vector defined in (8) verify :

$$z_k = \sum_{i \geq 1} T_i x_{k+1}^{[i]} \qquad \textbf{(11)}$$

and :

$$x_{k+1}^{[p]} = \sum_{i \geq 1} F_i^p x_k^{[i]}$$

with :

$$F_i^p = \sum_{j=1}^{i-n+1} (F_{i-j}^{p-1} \otimes F_j^1)$$

The solution of such nonlinear problem is developed in our previous works [11] and formulated by the main result given by the following relations :

$$vec(T_i) = -A_i^+ B_i + A_i^+ C_i vec(\alpha_i) \qquad \textbf{(12)}$$

$$vec(\alpha_i) = -(A_i^+ B_i)^+ A_i^+ C_i \qquad \textbf{(13)}$$

Where

$$A_i = [((R_n^i)^T \otimes M) - ((f_1 + g_0 \alpha_1)^{[i]} R_n^i)^T \otimes I_n]$$

$$B_i = vec((f_r + \sum_{j=1}^{r-1} g_j(\alpha_{i-j} \otimes I_{n^j}))R_n^i + T_2 F_i^2 R_n^i + \cdots + T_{i-1} F_i^{i-1} R_n^i)$$

$$C_i = ((R_n^i)^T \otimes g_0)$$

and $(A_i^+ B_i)^+$ is the Moore Penrose

generalized inverse of $(A_i^+ B_i)$.

3.2. Simulation results

The discretization of the continuousm odel of the synchronous generator whose characteristics are given in appendix C; leads to the polynomial form(5) with :

$$A_1 = \begin{bmatrix} 0.9099 & 0.9361 & -0.0227 \\ -0.1802 & 0.8721 & -0.0454 \\ 0.0219 & -0.2274 & 0.1149 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0.04 & 0.07 & -0.03 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.08 & 0.14 & -0.06 & 0 & 0 & 0 & 0 & 0 & 0 \\ -0.01 & -0.19 & 0.01 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} A_3(1,1) = 0.0176, A_3(1,2) = -0.023, A_3(1,3) = 0.0205 \\ A_3(2,1) = 0.0353, A_3(2,2) = -0.0459, A_3(2,3) = 0.0409 \\ A_3(3,1) = -0.0043, A_3(3,2) = 0.127, A_3(3,3) = -0.005 \\ A_3(i,j) = 0 \text{ for others } i,j, i = 1,\ldots,3 \text{ and } j = 1,\ldots,27 \end{cases}$$

$$B = \begin{bmatrix} -3.0828 \\ -6.1657 \\ 51.3866 \end{bmatrix}$$

The exploitation oft he control approach with the discrete model allows us to determine the polynomial control law :

$$u_k = \alpha_1 x_k + \alpha_2 x_k^{[2]} + \alpha_3 x_k^{[3]} \qquad \textbf{(14)}$$

and the non linear transformation :

$$z_k = x_k + T_2 x_k^{[2]} + T_3 x_k^{[3]} \qquad \textbf{(15)}$$

The matrix line α_1 is computed such that the dynamic of the linear system (9) will be characterized by the poles 0.6;0.7;0.7, which lead to :

$$\alpha_1 = [0.0086, -0.0034, -0.004]$$

Using equation(13), the matrices α_2 and α_3 becomes :

$$\alpha_2 = [0.01, 0.076, 0, 0.0076, 0.0039, 0, 0, 0, 0]$$

$$\begin{cases} \alpha_3(1,1) = -0.0065, \alpha_3(1,2) = -0.0024 \\ \alpha_3(1,4) = -0.0024, \alpha_3(1,5) = -0.0037 \\ \alpha_3(1,10) = -0.0024, \alpha_3(1,11) = -0.0037 \\ \alpha_3(1,13) = -0.0037, \alpha_3(1,14) = -0.0022 \\ \alpha_3(1,j) = 0 \text{ for others } j, j = 1,\ldots,27 \end{cases}$$

To valid the control law (14), a simulation study is conducted. Simulation results are shown in figure (2) where the evolution oft he variables $\Delta\delta_k$; $\Delta\omega_k$ and Δi_{f_k} are represented.

In these figures, it appears a visible improvement of the process performances; particularly from the view point of the rate oscillation damping and the rapid dynamics of the different variables of the machine. Such behavior is obtained by means of satisfactory and suitable control signal.

4. Conclusion

The study that we have presented in this work is carrying out a new approach of nonlinear discrete control of synchronous generator.

Thisa pproach is based on the linearisation of the controlled system via a nonlinear transformation of the state vector, leading to interesting results.

Such work exploits the conventional three dimensional reduced order model of a synchronous generator characterized by an important degree of non linearities. The application of the above method using high order generator modeling will be investigated later.

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Appendix A

$$p_1 = -\frac{u_s X_{ad}}{T_m (x_d + x_L)}$$

$$p_2 = \frac{0.5 u_s^2 (x_d - x_q^{'})}{T_M (x_q + x_L)(x_L + x_d)}$$

$$p_3 = \frac{1}{T_M} \; ; p_4 = -\frac{(x_L + x_d)}{(x_L + x_d^{'})\tau_{do}^{'}}$$

$$p_5 = -\frac{(x_d - x_d^{'}) u_s \omega_b}{(x_L + x_d^{'}) X_{ad}} \; ; p_6 = \frac{(x_L + x_d)}{\tau_{do}^{'} r_f (x_d^{'} + x_L)}$$

$$d_2 = -\tau_{qo}^{'}(1 - \frac{x_q^{'}}{x_q})\frac{u_s^2}{x_q}$$

Appendix B

$$a_1 = 2 p_2 c_2 + c_1 p_1 \xi_{n_3} \; ; a_2 = d_2 c_1^2$$

$$a_3 = s_1 p_1 \; ; a_4 = -2 p_2 s_2 - \frac{s_1 p_1 \xi_{n3}}{2}$$

$$a_5 = -2 d_2 c_1 s_1 \; ; a_6 = c_1 p_1$$

$$a_7 = -\frac{4}{3} c_2 p_2 - \frac{c_1 p_1 \xi_{n3}}{3!};$$

$$a_8 = s_1^2 - c_1^2$$

$$a_9 = -\frac{s_1 p_1}{2} \; ; b_1 = p_4 s_1 \; ; b_2 = p_5 \; ; b_3 = p_4 c_1$$

$$b_4 = -\frac{p_4 s_1}{2} \; ; s_1 = \sin(\xi_{n1}) \; ; c_1 = \cos(\xi_{n1})$$

$$s_2 = \sin(2\xi_{n1}) \; ; c_2 = \cos(2\xi_{n1})$$

Appendix C

$P_n = 360\,MW ; S_n = 432\,MVA ;$

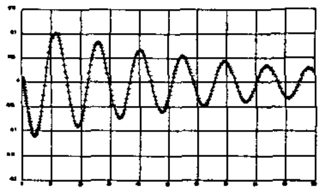
$x_L = .3\,p.u ; x_d = 2.459\,p.u, \omega_b = 1\,p.u$

$x_q = 2.354\,p.u ; x_d^{'} = .315\,p.u$

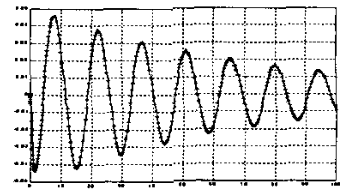
$\tau_{do} = 7.95 s ; T_M = 8 s ; X_{ad} = 2.28\,p.u$

$x_q^{'} = .476\,p.u ; \tau_{qo}^{'} = .39 s ; x_r = .191\,p.u$

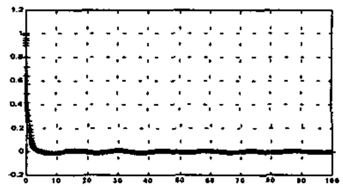
$r_f = .002\,p.u ; \tau_{do}^{''} = .01 s ; \tau_{qo}^{''} = .016 s.$



Evolution of the generator angle variation $\Delta\delta_k$

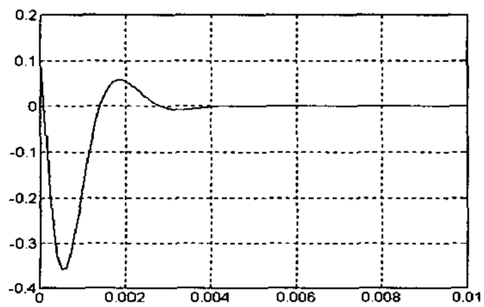


Evolution oft he generator angluar speed variation $\Delta\omega_k$

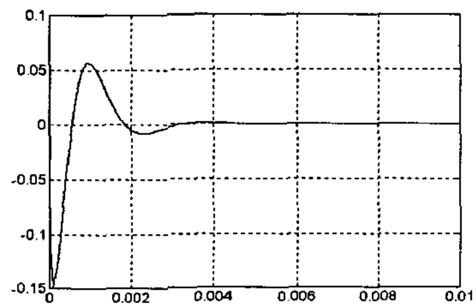


Evolution of the field winding current variation Δi_f

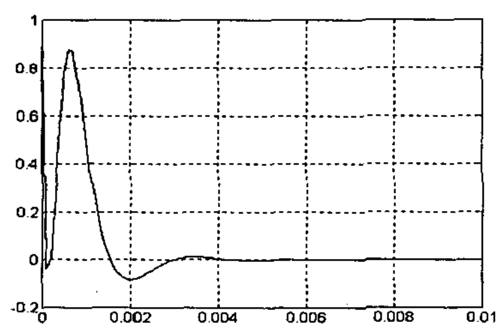
Figure (1) : Simulation of model (5) and (3)of the generator in free regime
(++Discrete model – Continuous model)



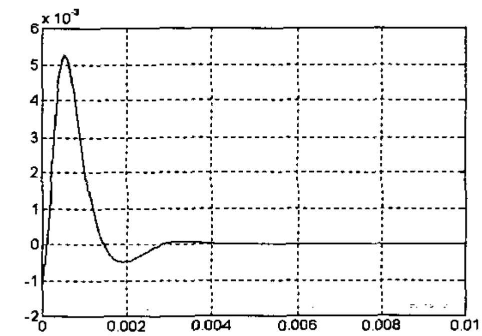
Variation of the generator angle $\Delta\delta_k$



Variation angular speed $\Delta\omega_k$



Variation of the field winding current Δi_{f_k}



Variation of the control Δu_f

Figure(2) : Simulation of the model of the generator provided by the control law (14)